

DS 203 – Big Data Essentials
Advanced SQL Lab 1 – Simple Views
Marks: 100

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Answer the following questions:

Q1. Create a view to show customer names and emails.

Query:

```
CREATE VIEW customer_email_view AS
SELECT first_name,
       last_name,
       email
FROM customer
```

Output Screenshot:

```
dvdrental=# select * from customer_email_view LIMIT 10;
 first_name | last_name | email
-----+-----+-----
Jared      | Ely      | jared.ely@sakilacustomer.org
Mary       | Smith    | mary.smith@sakilacustomer.org
Patricia   | Johnson  | patricia.johnson@sakilacustomer.org
Linda      | Williams | linda.williams@sakilacustomer.org
Barbara    | Jones    | barbara.jones@sakilacustomer.org
Elizabeth  | Brown    | elizabeth.brown@sakilacustomer.org
Jennifer   | Davis    | jennifer.davis@sakilacustomer.org
Maria      | Miller   | maria.miller@sakilacustomer.org
Susan      | Wilson   | susan.wilson@sakilacustomer.org
Margaret   | Moore    | margaret.moore@sakilacustomer.org
(10 rows)
```

Q2. Create a view to list film titles and their rental durations.

Query:

```
CREATE VIEW  
    film_rental_info  
AS  
SELECT  
    title,  
    rental_duration  
FROM  
    film;
```

Output Screenshot:

```
CREATE VIEW  
dvdrental=# SELECT * FROM film_rental_info LIMIT 10;  
          title          | rental_duration  
-----+-----  
Chamber Italian         |                7  
Grosse Wonderful        |                5  
Airport Pollock         |                6  
Bright Encounters       |                4  
Academy Dinosaur        |                6  
Ace Goldfinger          |                3  
Adaptation Holes        |                7  
Affair Prejudice        |                5  
African Egg             |                6  
Agent Truman            |                3  
(10 rows)
```

Q3. Query the view to find all customers with email ending in .org.

Query:

```
CREATE VIEW customer_email_org
AS
SELECT
    *
FROM
    customer_info
WHERE
    email
LIKE
    '%.org';
```

Output Screenshot:

```
dvdrental=# SELECT * FROM customer_email_org LIMIT 10;
 first_name | last_name | email
-----
Jared      | Ely       | jared.ely@sakilacustomer.org
Mary       | Smith     | mary.smith@sakilacustomer.org
Patricia   | Johnson   | patricia.johnson@sakilacustomer.org
Linda      | Williams  | linda.williams@sakilacustomer.org
Barbara    | Jones     | barbara.jones@sakilacustomer.org
Elizabeth  | Brown     | elizabeth.brown@sakilacustomer.org
Jennifer   | Davis     | jennifer.davis@sakilacustomer.org
Maria      | Miller    | maria.miller@sakilacustomer.org
Susan      | Wilson    | susan.wilson@sakilacustomer.org
Margaret   | Moore     | margaret.moore@sakilacustomer.org
(10 rows)
```

Q4. Drop the film_rental_info view.

Query:

```
DROP VIEW IF EXISTS film_rental_info;
```

Output Screenshot:

```
dvdrental=# DROP VIEW IF EXISTS film_rantal_info;
NOTICE: view "film_rantal_info" does not exist, skipping
DROP VIEW
```

Q5. Modify the customer view to include the full name (first name + last name) in one column.

Query:

```
DROP VIEW customer_info CASCADE;
CREATE VIEW customer_info
AS
SELECT
    (first_name || ' ' || last_name) AS full_name,
    email
FROM
    customer;
```

Output Screenshot:

```
dvdrental=# select * from customer_info LIMIT 10;
 full_name | email
-----+-----
 Jared Ely | jared.ely@sakilacustomer.org
 Mary Smith | mary.smith@sakilacustomer.org
 Patricia Johnson | patricia.johnson@sakilacustomer.org
 Linda Williams | linda.williams@sakilacustomer.org
 Barbara Jones | barbara.jones@sakilacustomer.org
 Elizabeth Brown | elizabeth.brown@sakilacustomer.org
 Jennifer Davis | jennifer.davis@sakilacustomer.org
 Maria Miller | maria.miller@sakilacustomer.org
 Susan Wilson | susan.wilson@sakilacustomer.org
 Margaret Moore | margaret.moore@sakilacustomer.org
(10 rows)
```

Marking Criteria:

For each question:

Correct Query = 10 Marks

Correct Output Screenshot = 10 Marks

Total: 20 Marks per question

Total Marks: 20 x 5 (questions) = 100 Marks.

Submission Instructions:

Submit the following:

1. Convert this document into PDF and rename it as “**DS 203 - Lab 1 – StudentID.pdf**” then submit in the appropriate submission on Google Classroom.
2. Also, submit the .sql script containing queries for all the questions in single file. You can name the script file as “**DS 203 - Lab 1 – StudentID.sql**”.